

COMMUNICATION LABORATORY

Experiment 5 : FM Demodulation



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GROUP 16 (Thursday Batch)

Introduction: Frequency Modulation (FM) involves reflecting the information about the required message signal in the carrier signal's frequency variation. To demodulate such signals and obtain the message signal back from the frequency information, we need to track how much the frequency varies from the central carrier frequency, which can be done via circuits known as Phase-Locked-Loops. Enabling us to relate frequency deviation to voltage, frequency demodulation is achieved, investigated here via the IC565.

Key Objectives:

1. Design and setup the IC565 frequency demodulation circuit.
2. Observe both time and frequency domain plots for the demodulated output.
3. Workout a schematic to demodulate square waves.
4. Demodulate the FM signal using a differentiator circuit with a DC Blocker.

Components Used:

1. Mixed Signal Oscilloscope
2. Breadboard
3. +/- 5V DC Power Supply
4. 5.6nF, 33nF, 10 μ F, 1nF and 0.1 μ F Capacitors
5. 560 Ω , 330 Ω , and 10k Ω resistors
6. 4.7k Ω Potentiometer
7. Function Generator
8. IC 565 Phase Locked Loop IC
9. LM741 Op-Amps
10. 1N4007 Silicon Diodes

Brief Theory:

A **Phase-Locked Loop (PLL)** is a special type of feedback control system that can automatically lock onto the frequency and phase of an incoming signal.

A PLL has three building blocks:

Phase Detector (PD):

This compares the phase of the incoming FM signal with the phase of the locally generated signal from the VCO. The output of the phase detector is an error signal that indicates how much the two signals differ in phase.

Low-Pass Filter (LPF):

The error signal from the phase detector usually contains both a useful low-frequency component and unwanted high-frequency components. The LPF smooths this signal, allowing only the useful error voltage to pass through while removing the high-frequency noise.

Voltage-Controlled Oscillator (VCO):

The VCO generates a signal whose frequency is not fixed but instead controlled by the input voltage (from the PD + LPF). This allows the VCO to continuously adjust itself so that it “tracks” the incoming FM signal.

Under lock:

- The VCO frequency follows the instantaneous frequency of the input FM wave.
- The control voltage (coming from the phase detector and filter) carries information about how much the input FM signal is deviating from the natural frequency of the VCO.
- When the PLL is locked, the VCO “chases” these frequency changes.
- The control voltage driving the VCO is directly proportional to the frequency deviation, and therefore proportional to the message signal $m(t)$.
- By taking this control voltage as the output, we effectively recover the original message signal.

Let the carrier be represented as:

$$v_c(t) = A_c \sin(2\pi f_c t)$$

The FM Wave can be written as:

FM Wave:

$$S(t) = A_c \cos \left(\omega_c t + K_f \int m(t) dt \right)$$

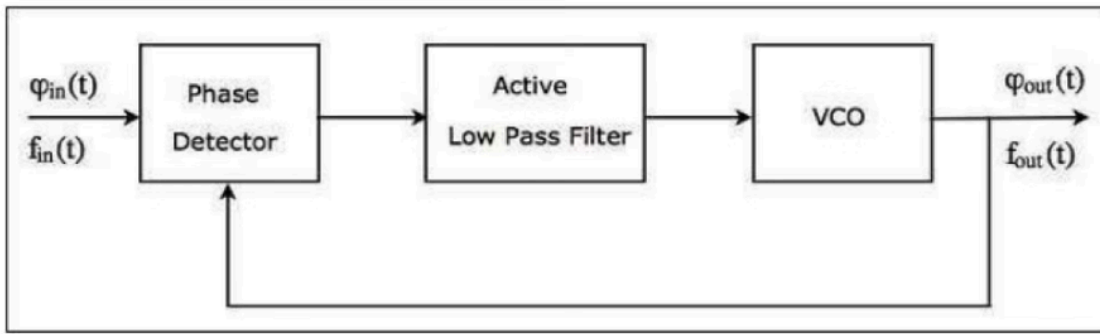
Where K_f = frequency sensitivity.

As can be seen, the frequency can be given as

$$\omega = \omega_c + K_f \cdot m(t)$$

The deviation from the central frequency follows the message signal, thus the above PLL setup works.

FM Demodulation based on PLL (Phase-Locked Loop):

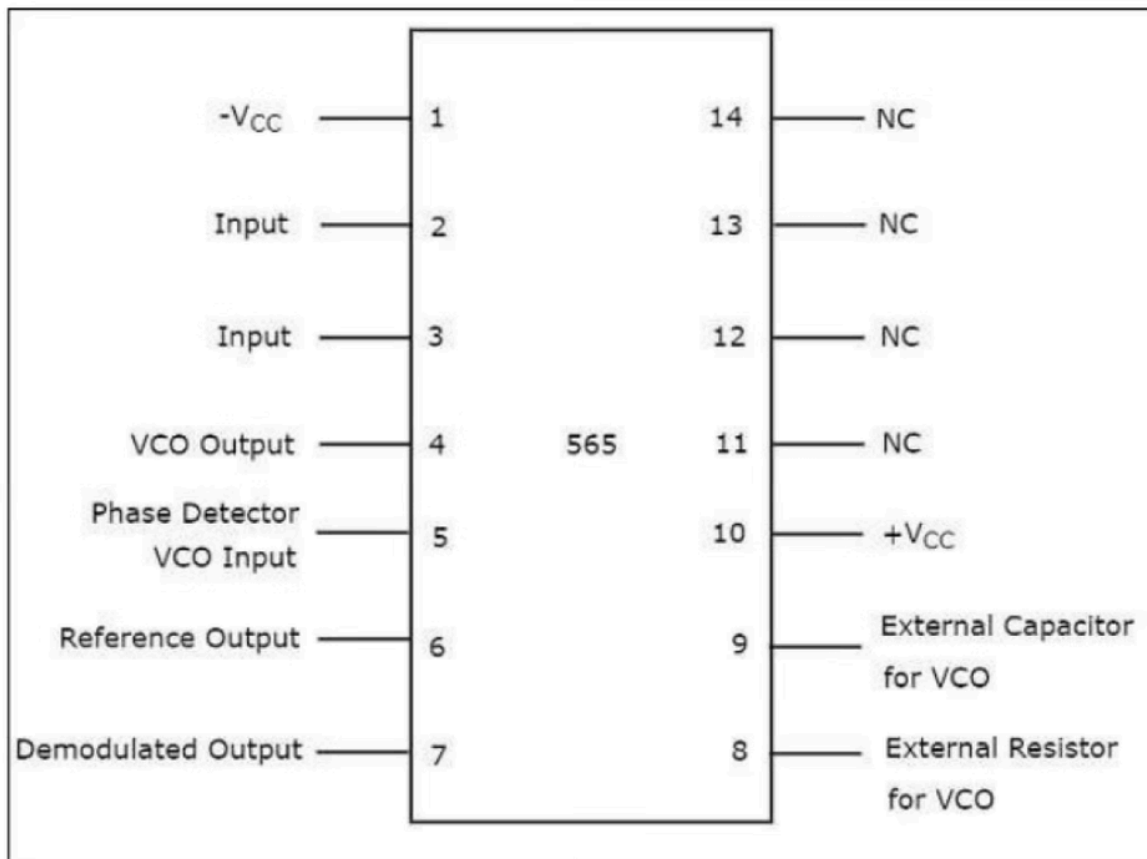


Note that a PLL is unable to track sudden signal changes. This means that a square wave cannot be demodulated this way. Thus, to demodulate, we use the fact that the low frequency components in the modulated signal correspond to the negative cycle of the square wave. Filtering those out means significantly lowering the modulated signal amplitude for the duration of the negative cycle. Thus, an envelope detector can track this amplitude variation, and with an aptly calibrated comparator, we can recover the signal with close to ideal duty-cycle, albeit not exactly accurate.

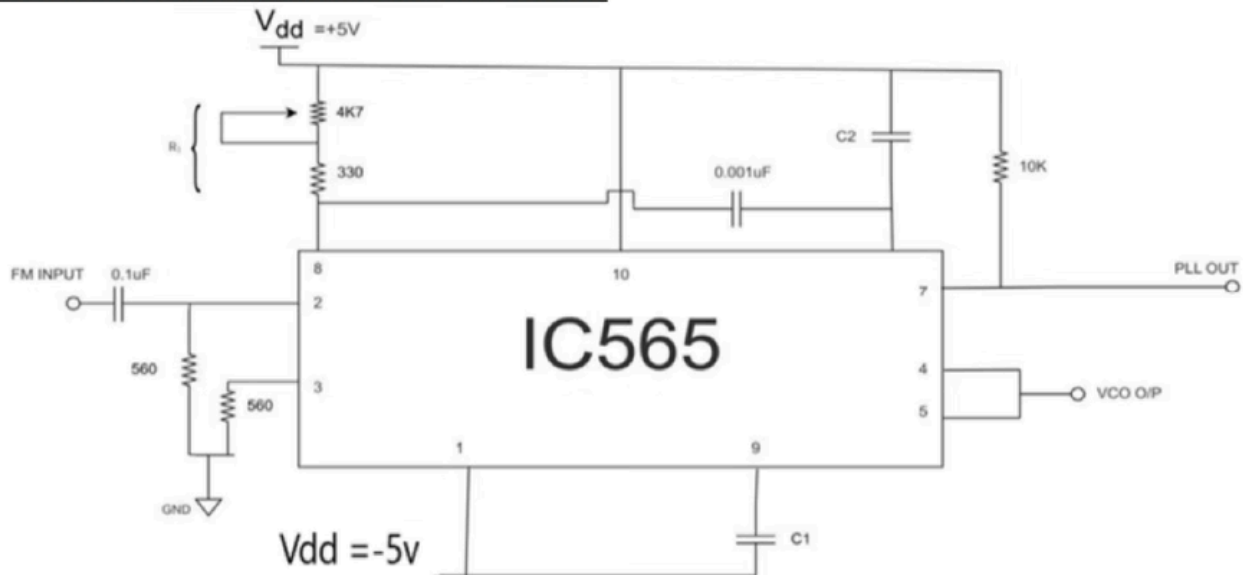
Yet another way to recover the message signal is simply via differentiation of the FM signal with respect to time. This results in the message signal modulating the amplitude of the resulting signal. Note that the resulting signal still has frequency variation according to the message signal, but the envelope is all we care about. Thus, we proceed as we did in AM Demodulation, using an envelope detector followed by a DC Blocker,

Circuit Design and Circuit Diagram:

IC 565 Pin diagram:



IC 565 Pin Connection with External Circuit



DESIGN:

Free running frequency of VCO,

Max= 12v to +12v Both I/P terminals (Pin 2 & 3) must receive identical DC bias range from 0 to 4V. R1 value is between $2K\Omega$ to $20K\Omega$. O/P Voltage swings -ve as I/P freq increases.

$$f_o = \frac{1}{4R_1C_1}$$

Filter (Internal):

Low pass filter

$$f = \frac{1}{2\pi \times 3.6 \times 10^3 \times C_2}$$

$$C_2 = 0.033 \mu F$$

$$f = \frac{1}{2 \times 3.141 \times 3.6 \times 10^3 \times .033 \times 10^{-6}}$$

$$= 1.3 \text{ KHz}$$

$$f_0 = 10 \text{ kHz}$$

From circuit:

- R_1 is adjustable (pot + resistor)
- Typically around $4.7 \text{ k}\Omega + 330 \Omega \approx 5 \text{ k}\Omega$ (approx)

$$C_1 = \frac{1}{4 \times 5 \times 10^3 \times 10^4}$$

$$C_1 = \frac{1}{2 \times 10^8} = 5 \times 10^{-9}$$

$$C_1 = 5 \text{ nF}$$

An additional low pass filter can be added with the circuit to eliminate frequency components other than modulating signal frequency 1 KHz from PLL o/p.

Design calculation of external Low Pass Filter

The message frequency is

$$f_m = 1 \text{ kHz}$$

We choose a cut off frequency :

$$f_c = 1.5 \text{ kHz}$$

$$R = 10 \text{ k}\Omega$$

Then capacitor becomes:

$$C = \frac{1}{2\pi R f_c}$$

Substitute:

$$C = \frac{1}{2\pi \times 10^4 \times 1.5 \times 10^3}$$

$$C = \frac{1}{94.2 \times 10^6}$$

$$C \approx 10.6 \text{ nF}$$

So the nearest standard value is:

$$C = 0.01 \mu F$$

RESULTS:

Without providing the FM Input, the internal VCO Output Waveform of PLL IC on the oscilloscope.
Expected :

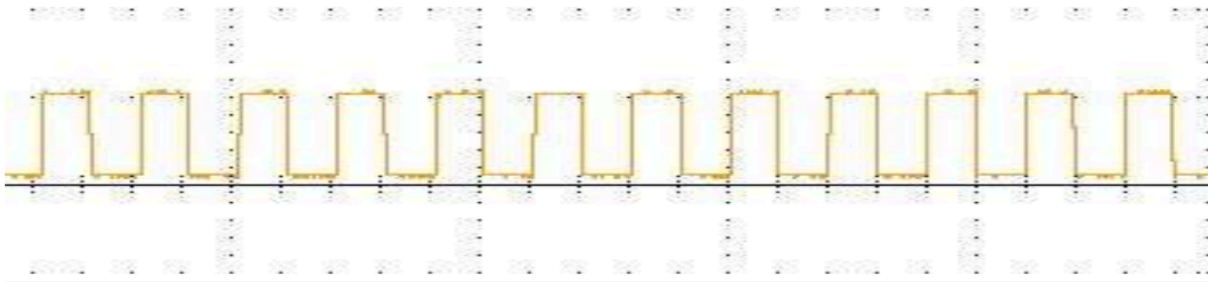


Fig -1. VCO Output Waveform of PLL IC

Received :

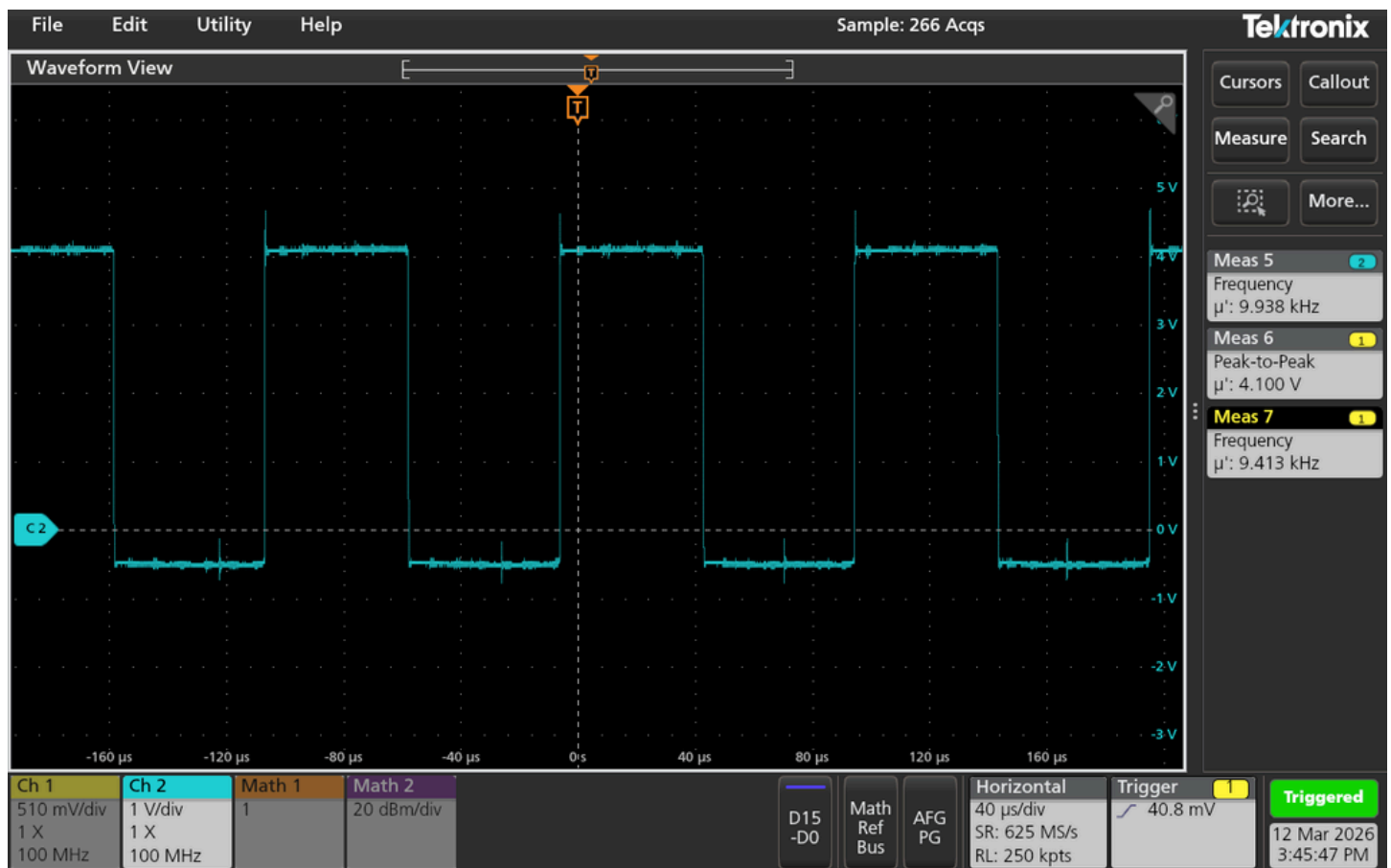
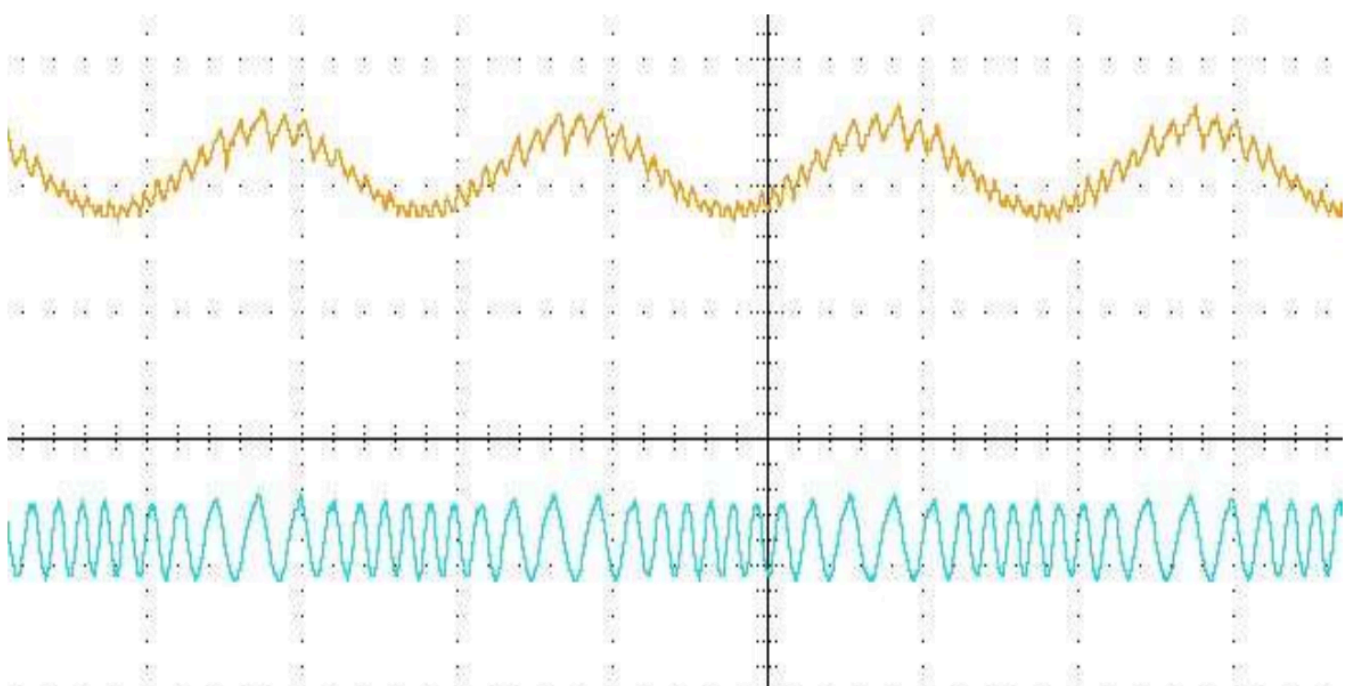


Figure: VCO Output Waveform of PLL IC

Adjust the potentiometer at R1, to set the frequency of VCO O/p to 10 KHz (10 KHz is the carrier frequency chosen for modulation). The FM Input is given at pin 2 of PLL IC and the PLL O/p waveform is observed at pin no 7 of IC.

Expected:



Received :

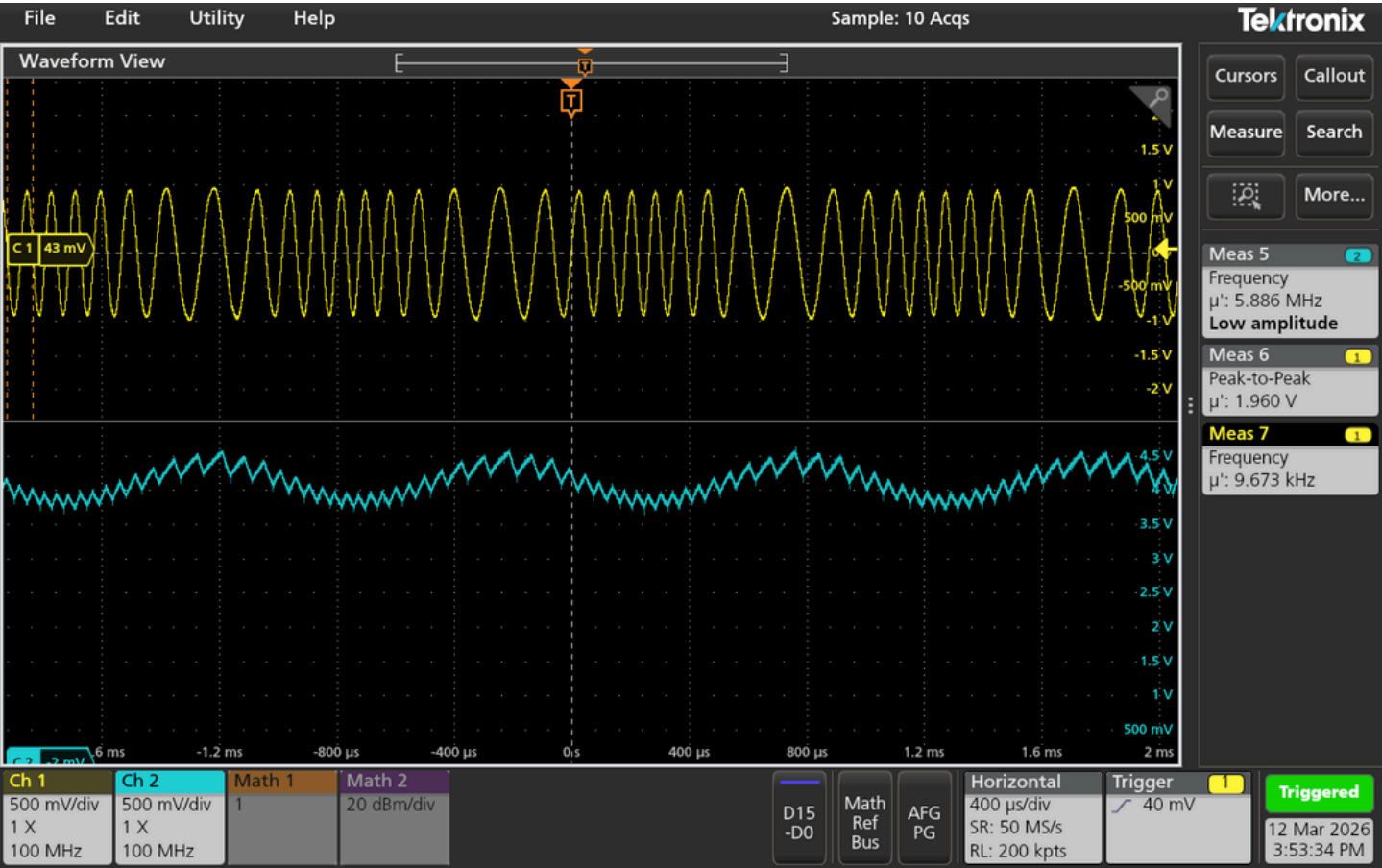


Figure: (a) FM Input Waveform (b) PLL Output waveform (Demodulated signal)

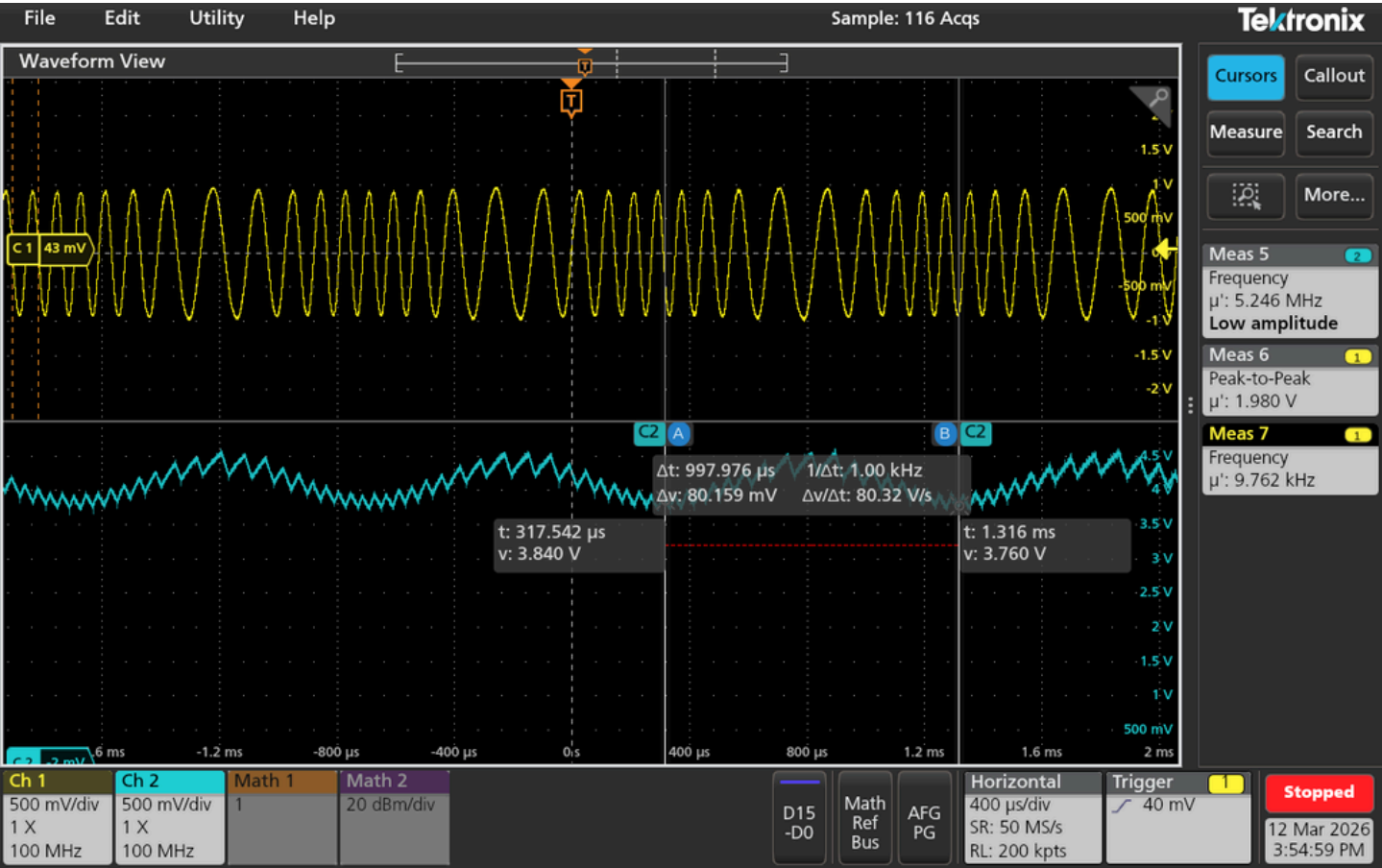


Figure: Measuring $f_m = 1\text{KHz}$

DISCUSSION :

The experiment demonstrates the use of a Phase-Locked Loop (PLL) with an internal Voltage Controlled Oscillator (VCO) for frequency demodulation of FM signals. The performance of this method depends strongly on frequency synchronization, loop dynamics, and signal characteristics.

1. Effect of Frequency Mismatch

A key observation is that the quality of the recovered message signal deteriorates significantly when the VCO free-running frequency does not match the carrier frequency f_c .

- The PLL tracks frequency deviations relative to its own center frequency.
- Any mismatch shifts the operating point, leading to:
 - Distortion in the demodulated waveform
 - Asymmetry in sinusoidal outputs
- This occurs because the PLL interprets deviations incorrectly when it is not locked precisely to the carrier.

2. Phase Shift in Demodulated Output

The recovered signal exhibits a phase shift relative to the original message.

- FM signals encode information in frequency (i.e., derivative of phase).
- The PLL demodulation process inherently introduces:
 - Phase delay due to loop dynamics
 - Additional delay from filters and internal components
- The VCO itself contributes phase offsets that may vary with temperature and supply voltage, as indicated in device characteristics.

3. Behavior of Output Signal

It was observed that:

- Maximum amplitude in the demodulated signal corresponds to minimum instantaneous frequency (rarefactions).
- Minimum amplitude corresponds to maximum instantaneous frequency (compressions).

This inversion-like behavior is a direct consequence of how the PLL control voltage responds to frequency deviations.

4. Limitations with Square Wave Demodulation

PLL-based demodulation is not ideal for square wave signals:

- Square waves cause abrupt frequency and phase transitions in the FM signal.
- The PLL cannot track these rapid changes in real time due to finite loop bandwidth.

As a result:

- The recovered waveform shows:
 - Increased rise and fall times
 - Duty cycle variations ($\approx \pm 10\%$)
- This makes the method unreliable for accurate square wave recovery.

5. DC Offset and Noise Effects

The demodulated output contains a noticeable DC offset, typically:

- Originating from the VCO output bias ($\approx 4\text{--}4.5\text{ V}$ as per datasheet behavior)
- This can be removed using:
 - Capacitive coupling (high-pass filtering)

Additionally:

- Low-frequency noise (e.g., 50 Hz interference) can introduce slow variations.
- Proper filtering (e.g., band-pass filters around the message frequency) helps improve signal quality.

6. Role of Buffering and Circuit Design

Loading effects between stages were found to degrade signal integrity.

- Use of unity gain buffers between stages:
 - Prevents impedance mismatch
 - Maintains waveform shape
- Active filters can further improve performance by:
 - Reducing attenuation
 - Providing better control over frequency response

However, these improvements increase circuit complexity.

7. Overall Performance and Applications

PLL-based FM demodulation is effective for:

- Sinusoidal message signals
- Situations where:
 - Carrier frequency is stable
 - Proper lock condition is maintained

Typical applications include:

- FM communication receivers
- Telemetry systems

- Analog signal recovery in communication circuits